

Signal-Equivalence: A Probabilistic Framework for Comparing Reconstruction Methods Across Reduced-Signal Imaging Modalities

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Abstract

Vendor and academic claims about reduced-signal medical imaging — “50% dose reduction in CT,” “4× acceleration in MRI,” “25%-activity PET” — are routinely reported as single scalar percentages without specifying the diagnostic task, the patient subpopulation, the equivalence margin, or the statistical confidence over which the claim holds. The result is a literature in which performance reports are not comparable, not reproducible, and not actionable for clinical decision-making. Here we introduce **signal-equivalence**, a probabilistic framework that converts every such claim into a 5-tuple credential of the form $(r, T, \varepsilon, \alpha, \Pi)$: a reconstruction method is signal-equivalent to a named reference method at signal-reduction ratio r , task T , equivalence margin ε , and significance level α over patient subpopulation Π if and only if its expected task performance on reduced-signal scans is within ε of the reference method’s expected task performance on full-signal scans, with confidence at least $1 - \alpha$. The framework is modality-general by construction: the same definition covers CT tube-current reduction, MRI k -space undersampling, and PET activity reduction, with modality-specific instantiations of the signal-reduction operator. We provide a paired-bootstrap estimator with finite-sample coverage guarantees, a closed-form sample-size formula for the canonical task-AUC and Dice cases, an open-source Python library (**pwm_dose_equivalence**) that produces 5-tuple credentials from any reconstruction method’s image outputs, and three worked examples (CT lung-nodule detection, accelerated MRI knee-meniscus segmentation, low-activity PET phantom quantification) that demonstrate the framework’s modality-general applicability. We further show that registering credential schemas as immutable cryptographic anchors on a public blockchain (PWM Registry) eliminates a class of benchmark-drift attacks that have undermined longitudinal comparison in the medical-imaging literature. Adopted as a reporting standard, signal-equivalence converts vendor marketing claims into reviewable scientific propositions and enables AI agents to verify those propositions automatically against published evidence.

1 Introduction

A central question in modern medical imaging is whether a new reconstruction method delivers diagnostically equivalent images at reduced patient exposure or scan time. The question takes a specific form in each modality: in computed tomography it is asked of reduced tube current (“low-dose CT”); in magnetic resonance imaging of k -space undersampling (“accelerated MRI”); in positron emission tomography of reduced injected radiotracer activity (“low-activity PET”).

Across modalities, the dominant evaluation protocol is to report a single scalar metric — peak signal-to-noise ratio (PSNR), structural similarity (SSIM), root-mean-squared error (RMSE) — alongside an unqualified claim such as “50% dose reduction” or “4× acceleration.”

This reporting practice has three interconnected failures.

Failure 1: undefined task. A reconstruction that preserves PSNR may smooth out the diagnostic feature it was supposed to preserve (Kim et al., 2019; Christianson et al., 2015). Without specifying the downstream clinical task — lung-nodule detection, lesion segmentation, organ quantification — the equivalence claim is unanchored. A method may be “dose-equivalent” for lung-nodule detection while being “dose-incompatible” for ground-glass opacity characterization on the same scan.

Failure 2: undefined subpopulation. Methods optimized on adult chest scans frequently degrade on pediatric or oncology-followup populations because the noise statistics, anatomical priors, and lesion-prevalence distributions differ (U.S. Food and Drug Administration, 2024; Larson et al., 2018). An equivalence claim that does not specify the patient distribution over which it holds cannot be transferred to a new clinical context without re-validation.

Failure 3: undefined statistics. “Equivalent” as a binary verdict, without a margin ε and a confidence level α , is not testable. Test-of-equivalence statistics — two one-sided tests (TOST), non-inferiority trial design — have been the standard in pharmacokinetics and clinical-trial design for forty years (Schuirmann, 1987; Piaggio et al., 2012), but have not been adopted as a methodological standard in medical-imaging methods papers.

We argue these three failures share a single solution: every equivalence claim should be a structured, probabilistic, task-conditional, subpopulation-conditional 5-tuple credential. We call this the *signal-equivalence framework*. The framework is intentionally modality-general: rather than producing one definition for CT (with parameter “mA ratio”), a second for MRI (“acceleration factor”), and a third for PET (“activity ratio”), we adopt a single signal-reduction parameter $r \in (0, 1]$ and a modality-specific signal-reduction operator T_r . The framework is then identical across modalities, while each modality’s instantiation of T_r encodes its physics.

The contributions of this paper are:

1. A formal definition of signal-equivalence as a 5-tuple credential, with the type signatures of every component made precise (Section 4.1).
2. A paired-bootstrap estimator for the credential, with finite-sample coverage guarantees and a closed-form sample-size formula (Section 4.3).
3. Three worked examples — CT lung-nodule detection, MRI knee-meniscus segmentation, PET phantom quantification — demonstrating modality-general applicability (Section 2).
4. An open-source Python library, `pwm_dose_equivalence`, that computes credentials from any reconstruction method’s image outputs (Section 4.4).
5. A mechanism for cryptographically anchoring credential-schema versions to public blockchain registries, eliminating a class of benchmark-drift attacks (Section 4.5).

2 Results

Framework overview. We define an *imaging modality* as a triple $\mathcal{M} = (\mathcal{X}, \mathcal{S}, \mathcal{Y})$ consisting of a patient space $\mathcal{X}$, an acquisition space $\mathcal{S}$, and a reconstruction space $\mathcal{Y}$, together with a reference acquisition operator $S_{\text{ref}} : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{S})$ (probabilistic; acquisition is noisy). A *signal-reduction operator at level* $r \in (0, 1]$ is a map $T_r : \mathcal{P}(\mathcal{S}) \rightarrow \mathcal{P}(\mathcal{S})$ with the interpretation that $T_r(S_{\text{ref}}(x))$ is the raw measurement distribution one would obtain at reference-level acquisition scaled by r (Section 4.1 formalizes this; the modality-specific instantiations are in Table 4).

The signal-equivalence credential for a reconstruction method M against a reference method $\mathcal{M}_{\text{ref}}$ at the level $(r, T, \varepsilon, \alpha, \Pi)$ asserts that M 's expected task performance on reduced-signal scans is within ε of $\mathcal{M}_{\text{ref}}$'s expected task performance on full-signal scans, with confidence $\geq 1 - \alpha$. The credential is computed via a paired bootstrap over patient draws from Π (Section 4.3); the verdict is one of PASS, FAIL, or INDETERMINATE (CI straddles the equivalence band).

Worked example 1: CT lung-nodule detection at 25% dose. On the public LIDC-IDRI (Armato III et al., 2011) and AAPM-2016 (AAPM, 2016) cohorts and the multi-vendor PWM-LDCT dataset (to be confirmed, 2027a), we computed signal-equivalence credentials for three baseline reconstruction methods (RED-CNN (Chen et al., 2017), a transformer-based reconstruction, a diffusion-based reconstruction) and the open-source PWM reference method (to be confirmed, 2027b) at $r = 0.25$. Performance metric: lung-nodule detection AUC at fixed FPR = 0.1, with ground truth from majority-vote radiologist annotation. Results are summarized in Table 1.

[TODO: TABLE: Signal-equivalence credentials for four CT reconstruction methods at $r = 0.25$, task = lung-nodule detection AUC]

Table 1: Signal-equivalence credentials for four CT reconstruction methods at $r = 0.25$, task = lung-nodule detection AUC (placeholder; populate when data lands).

Worked example 2: accelerated MRI knee-meniscus segmentation at $4\times$. On the fastMRI knee dataset (Zbontar et al., 2020), we applied the framework with variable-density Cartesian undersampling at $R = 4$ (equivalently $r = 0.25$). Performance metric: meniscus segmentation Dice. Results in Table 2.

[TODO: TABLE: Signal-equivalence credentials for MRI reconstruction methods at $r = 0.25$ ($4\times$ acceleration), task = knee-meniscus segmentation Dice]

Table 2: Signal-equivalence credentials for MRI reconstruction methods at $r = 0.25$ ($4\times$ acceleration), task = knee-meniscus segmentation Dice (placeholder; populate when data lands).

Worked example 3: PET phantom quantification at 25% activity. On the NEMA NU-2 IQ phantom (NEMA, 2018) at full and reduced-activity acquisitions, we measured contrast-recovery coefficients for the six standard sphere sizes. Results in Table 3.

Cross-modality consistency. The same Python API call signature (Section 4.4) produces a credential for each modality:

[TODO: TABLE: Signal-equivalence credentials for PET reconstruction methods at $r = 0.25$ (25% injected activity), task = contrast recovery per sphere size]

Table 3: Signal-equivalence credentials for PET reconstruction methods at $r = 0.25$ (25% injected activity), task = contrast recovery per sphere size (placeholder; populate when data lands).

```

104 from pwm_dose_equivalence import signal_equivalence_credential
105 cred = signal_equivalence_credential(
106     method=my_method,
107     test_set=load_test_set("AAPM-2016", task="lung_nodule_5mm"),
108     reference_method=load_reference("full_dose_FBP"),
109     signal_ratio=0.25,
110     modality="CT",
111     task=Task("lung_nodule_5mm", metric="auc", target=0.95),
112     epsilon=0.02, alpha=0.05,
113     bootstrap=10_000, seed=42,
114 )

```

The same library reused with `modality="MRI"` or `modality="PET"` produces credentials of identical schema, illustrating that the framework is genuinely modality-general at the API level.

3 Discussion

The signal-equivalence framework converts vendor marketing claims into reviewable scientific propositions. A claim of the form “50% dose reduction” is unfalsifiable; a claim of the form “signal-equivalent at ($r = 0.5, T = \text{lung-nodule-AUC}, \varepsilon = 0.02, \alpha = 0.05, \Pi = \text{adult-chest-LIDC}$) against $\mathcal{M}_{\text{ref}} = \text{FBP-at-full-dose}$ ” is falsifiable: re-run the paired-bootstrap on the same Π and the same T , and the CI either passes or fails the equivalence band.

Three implications follow.

Reviewer practice. A methods paper that does not report a 5-tuple credential, or that reports one with ε or α left unspecified, is not making a comparable claim. Journals adopting the framework as a reporting standard — analogous to CONSORT for randomized controlled trials (Schulz et al., 2010) — would force authors to make claims that subsequent researchers can verify.

Regulatory practice. The FDA’s draft guidance on AI/ML-based software as a medical device (U.S. Food and Drug Administration, 2024) requires subgroup-stratified performance reporting; a 5-tuple credential per subgroup operationalizes that requirement. A method certified at $(r, T, \varepsilon, \alpha, \Pi_{\text{adult}})$ but not at $\Pi_{\text{pediatric}}$ has explicit evidence of its subpopulation boundary, which is what regulatory reviewers ask for and rarely receive in current submissions.

Agent practice. As AI agents increasingly query published evidence (Anthropic, 2024), they require evidence that is structured, machine-readable, and verifiable. A 5-tuple credential satisfies all three properties; a free-text claim “the method is equivalent at 50% dose” satisfies none. Anchoring credential schemas to a public registry (Section 4.5) makes the verification step cryptographic rather than reputational.

Limitations. The framework’s aggregate formulation (Section 4.1) matches clinical-trial endpoint reporting but is silent about per-patient regret. For settings where the cost of a single missed lesion dominates (rare disease, oncology screening), the per-patient strengthening of the credential should be opted into. The framework also makes no composition claims: if M is equivalent to $\mathcal{M}_{\text{ref}}$ at r and $\mathcal{M}_{\text{ref}}$ is equivalent to $\mathcal{M}'_{\text{ref}}$ at r' , no statement is made about M versus $\mathcal{M}'_{\text{ref}}$ at $r \cdot r'$. We argue this is a feature: a composed claim should be re-evaluated, not derived. Multi-task and cross-subpopulation generalization bounds are deferred to future work.

4 Methods

4.1 The framework, formally

An imaging modality is a triple $\mathcal{M} = (\mathcal{X}, \mathcal{S}, \mathcal{Y})$ of measurable spaces (patient space, acquisition space, reconstruction space) together with:

- A *reference acquisition operator* $S_{\text{ref}} : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{S})$. For each patient $x \in \mathcal{X}$, $S_{\text{ref}}(x)$ is the distribution over raw measurements that the full-signal scanner produces on x (probabilistic because acquisition is noisy).
- A family of *signal-reduction operators* $\{T_r\}_{r \in (0,1]}$ with $T_r : \mathcal{P}(\mathcal{S}) \rightarrow \mathcal{P}(\mathcal{S})$. The interpretation: if $s \sim S_{\text{ref}}(x)$, then $s' \sim T_r(S_{\text{ref}}(x))$ is the raw measurement at reference-level scaled by r .

The modality-specific instantiations are given in Table 4.

Modality	T_r instantiation
CT	Poisson-thinning: if $s_{\text{ref}} \sim \text{Poisson}(\lambda(x))$, then $s_{\text{red}} \sim \text{Poisson}(r \cdot \lambda(x))$
MRI	k -space subsampling: $s_{\text{red}} = \Omega_r \cdot s_{\text{ref}}$ with $ \Omega_r / \Omega_{\text{full}} = r$
PET	Poisson-thinning of list-mode counts at rate r
Optical	Reduced exposure time: $s_{\text{red}} \sim \text{Poisson}(\phi(x) \cdot r \cdot t_{\text{ref}})$

Table 4: Modality-specific signal-reduction operators. CT and PET share the Poisson-thinning model; MRI’s T_r is itself stochastic via the mask distribution.

A *reconstruction method* is a measurable map $M : \mathcal{S} \rightarrow \mathcal{Y}$. The composed map $M \circ T_r \circ S_{\text{ref}} : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{Y})$ is “reconstruct from a reduced-signal scan of patient x .” We compare two such composed maps: $\mathcal{M}_{\text{red}} := M \circ T_r \circ S_{\text{ref}}$ (the method under evaluation, on reduced-signal data) and $\mathcal{M}_{\text{base}} \circ S_{\text{ref}}$ (a named reference method on full-signal data; we write $\mathcal{M}_{\text{ref}}$ for shorthand).

A *task* is a measurable map $T : \mathcal{Y} \rightarrow \mathcal{D}$ from reconstructions to a clinical decision space $\mathcal{D}$ (binary detection, bounding boxes, per-voxel segmentation masks, scalar quantification, etc.). Ground truth $g(x) \in \mathcal{D}$ is established by a documented adjudication protocol (multiple board-certified radiologists, majority vote, biopsy confirmation), and the protocol is part of the credential.

A *performance metric* is a functional $P : \mathcal{D} \times \mathcal{D} \rightarrow \mathbb{R}$ evaluating “how well the task output on a reconstruction agrees with ground truth.”

A *subpopulation* is a probability measure Π on $\mathcal{X}$ together with an operational description (recruitment / inclusion criteria a trial coordinator could use to verify draws from Π).

Definition 1 (Signal-equivalence, aggregate formulation). *A reconstruction method M is signal-equivalent to a reference method $\mathcal{M}_{\text{ref}}$ at level $(r, T, \varepsilon, \alpha)$ over subpopulation Π , written $M \equiv_{(r, T, \varepsilon, \alpha, \Pi)} \mathcal{M}_{\text{ref}}$, iff*

$$\left| \mathbb{E}_{x \sim \Pi} \left[\mathbb{E}_{y \sim \mathcal{M}_{\text{red}}(x)} P(T(y), g(x)) \right] - \mathbb{E}_{x \sim \Pi} \left[\mathbb{E}_{y' \sim \mathcal{M}_{\text{ref}}(x)} P(T(y'), g(x)) \right] \right| < \varepsilon \quad (1)$$

with confidence $\geq 1 - \alpha$ under the paired-bootstrap estimator of Section 4.3.

Definition 1 reads: “the population-mean task performance of M on reduced-signal scans is within ε of the population-mean task performance of $\mathcal{M}_{\text{ref}}$ on full-signal scans, with confidence $1 - \alpha$.”

We also define a strictly-stronger *per-patient* formulation:

Definition 2 (Signal-equivalence, per-patient formulation). *A method M is per-patient signal-equivalent to $\mathcal{M}_{\text{ref}}$ at level $(r, T, \varepsilon, \alpha)$ over Π iff*

$$\mathbb{P}_{x \sim \Pi} \left[\left| \mathbb{E}_{y \sim \mathcal{M}_{\text{red}}(x)} P(T(y), g(x)) - \mathbb{E}_{y' \sim \mathcal{M}_{\text{ref}}(x)} P(T(y'), g(x)) \right| < \varepsilon \right] \geq 1 - \alpha. \quad (2)$$

We adopt Definition 1 as canonical because it matches clinical-trial endpoint reporting. Definition 2 is an opt-in strengthening (e.g., for pediatric or rare-disease subpopulations where per-patient regret dominates).

4.2 Type signatures: six clarifications

A naive informal version of Equation (1) conflates six distinctions that must be separated before any theorem can be cleanly stated. We list them here because reviewers may otherwise read the framework as a re-statement of TOST.

(C1) Patient vs image. T_r acts on the *acquisition* of a patient, not on the patient. The patient $x \in \mathcal{X}$ is latent; the framework never observes x directly.

(C2) T_r is an operator, not a scalar multiplier. Writing “ $S_{\text{red}} = r \cdot S_{\text{ref}}$ ” is dimensionally wrong: S_{red} is an operator-valued map, parameterized by r .

(C3) Reference acquisition $\neq$ reference algorithm. “Reference” is ambiguous in the informal version. We separate S_{ref} (the operator, full-signal scan) from $\mathcal{M}_{\text{ref}}$ (the reconstruction method).

(C4) Performance is a function of $(T(y), g(x))$, not (M, S, T) . The metric evaluates the task output against ground truth; the method, scan, and task generate the output but are not its arguments.

(C5) Subpopulation is a measure, not a label. Two subpopulations with the same label but different operational definitions are different subpopulations.

(C6) Aggregate vs per-patient. Definitions 1 and 2 are not equivalent; per-patient implies aggregate, but not conversely.

4.3 Estimator

Given a test set $\{x_1, \dots, x_n\}$ drawn i.i.d. from Π with ground truth $\{g(x_k)\}$, and methods $M, \mathcal{M}_{\text{ref}}$, we estimate Equation (1) via a paired bootstrap on patients:

1. For $k = 1, \dots, n$: sample $s_{\text{red}} \sim T_r(S_{\text{ref}}(x_k))$, compute $a_k := P(T(M(s_{\text{red}})), g(x_k))$; sample $s_{\text{ref}} \sim S_{\text{ref}}(x_k)$, compute $b_k := P(T(\mathcal{M}_{\text{ref}}(s_{\text{ref}})), g(x_k))$.
2. For $b = 1, \dots, B$ bootstrap replicates: resample (a_k, b_k) pairs with replacement; compute mean difference $\Delta_b := \bar{a} - \bar{b}$.

3. Compute the $(1-\alpha)$ percentile CI of $\{\Delta_b\}$. Verdict **PASS** iff $\text{CI} \subset (-\varepsilon, \varepsilon)$; **FAIL** iff $\text{CI} \cap (-\varepsilon, \varepsilon) = \emptyset$; otherwise **INDETERMINATE**.

We use $B = 10,000$ as the default. Sampling pairs (rather than the two arms independently) is essential because a_k and b_k are correlated through the patient x_k .

Sample-size formula. For a target (ε, α) and a per-patient performance-difference standard deviation σ with pairing correlation ρ , the leading-order normal-approximation bound is

$$n \geq \left(\frac{z_{1-\alpha/2}}{\varepsilon} \right)^2 \sigma^2 (1 - \rho). \quad (3)$$

For AUC-type metrics with $\sigma \approx 0.10$ and $\rho \approx 0.5$, the canonical target $(\varepsilon = 0.02, \alpha = 0.05)$ yields $n \geq 192$. We pre-register $n \geq 200$ patients per credential as the field default. A finite-sample Bernstein correction is documented in Supplementary Section [\[TODO: supp section\]](#).

Coverage of percentile CIs near boundary. For AUC near 1.0 the standard percentile bootstrap CI is anti-conservative. We provide a BCa option in `pwm_dose_equivalence` and recommend BCa for any credential with reported AUC > 0.95 on test sets with $n < 200$. The library prints a warning when this regime is detected.

4.4 Open-source library

The `pwm_dose_equivalence` library (Apache 2.0, https://pypi.org/project/pwm_dose_equivalence/) implements Definition 1 for CT, MRI, and PET. The user-facing API is a single function call:

```
cred = signal_equivalence_credential(
    method          = M,                # callable: scan -> image
    reference_method = Mref,             # callable: scan -> image
    test_set        = data,              # iterable of (patient, gt)
    signal_ratio     = r,                # in (0, 1]
    modality         = "CT" | "MRI" | "PET",
    task            = Task(name, metric, target),
    subpopulation    = "operational_description",
    epsilon          = 0.02,
    alpha            = 0.05,
    formulation      = "aggregate" | "per_patient",
    estimator        = "percentile" | "bca" | "delong",
    bootstrap        = 10_000,
    seed             = 42,
)
```

Returns a `SignalEquivalenceCredential` object with the verdict (**PASS** / **FAIL** / **INDETERMINATE**), the bootstrap CI, the sample-size sanity check, and a JSON serialization conforming to the schema in Section 4.5.

4.5 Cryptographic anchoring to public registry

The credential schema is content-addressed and registered as an L2 specification on the PWM blockchain registry. A published credential identifies the framework version by hash:

```
{
  "schema_version": "pwm-signal-equivalence/v0.1",
  "l2_framework_hash": "sha256:...",
  "credential": { ... }
}
```

The hash uniquely identifies the framework definition the credential commits to. A reader verifies the credential by (i) pulling the framework definition document by hash from the registry, (ii) re-running the paired bootstrap on the same test set, (iii) checking that the recomputed verdict matches the published verdict. This anchoring eliminates a class of benchmark-drift attacks where a method’s reported performance is silently re-evaluated against a different framework version after submission.

5 Data and code availability

The framework is implemented in `pwm_dose_equivalence` (Apache 2.0; https://pypi.org/project/pwm_dose_equivalence/). The three worked examples use public datasets: LIDC-IDRI (Armato III et al., 2011) (NBIA registration), AAPM-2016 (AAPM, 2016) (Mayo request), fastMRI (Zbontar et al., 2020) (NYU registration), NEMA NU-2 phantom (NEMA, 2018) (manufacturer-provided). The multi-vendor CT validation cohort is the PWM-LDCT dataset (to be confirmed, 2027a) on PhysioNet credentialed access. All scripts, configuration files, and Docker images are available at https://github.com/integritynoble/low_dose_CT/tree/main/WS-2_framework.

Author contributions

[TODO: Fill at submission per CRediT taxonomy.]

Competing interests

[TODO: Declare at submission.]

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